Raza Shaud

Electrical Engineer

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Electrical Engineering graduate (June 2026) with expertise in power systems, machine learning, and industrial automation. Delivered 4 technical projects with 96.5% average accuracy, automating 99% of manual processes. Contributed to 3 industrial internships across power generation and steel manufacturing sectors. Published 2 research papers including IEEE Best Paper Award winner.

Education

Ghani Khan Choudhury Institute of Engineering and Technology, Malda, India *Oct 2022 – June 2026*
Bachelor of Technology in Electrical Engineering CGPA: 7.4/10

Projects

Solar-Powered Robotic Cable Monitoring System | Arduino, Solar PV, CNN, OpenCV, Python

- Engineered autonomous mobile robot integrating 12V solar PV system with camera-based cable insulation monitoring across 500m test infrastructure, eliminating 100% manual inspection requirements
- Architected CNN-based fault detection model processing 1300+ cable images achieving 96.5% accuracy, reducing false positives by 78% compared to traditional visual inspection methods
- Optimized power consumption achieving 8-hour continuous operation on 20W solar panel, cutting operational costs by ₹15,000 annually per monitoring unit

Human vs AI Voice Classification System | Python, Random Forest, MFCC

- Developed binary classification system analyzing 1500+ audio samples using MFCC feature extraction to distinguish authentic human voice from AI-generated speech with 89% accuracy
- Implemented Random Forest model with 15-dimensional feature vector, outperforming baseline SVM by 12% through hyperparameter tuning and cross-validation across 5 folds
- Preprocessed audio dataset reducing noise interference by 65% using spectral gating techniques, improving model generalization on real-world recordings

Face Recognition Attendance System | Python, OpenCV, Pandas

- Built real-time facial recognition attendance system processing 30 faces per second, automating attendance logging for 100+ students and reducing manual record-keeping by 99%
- Integrated face encoding database with Pandas-based CSV logging achieving 95% recognition accuracy under varying lighting conditions across 200+ daily scans
- Decreased attendance processing time from 15 min to 45 sec per session, saving 28 instructor-hours per semester

Saline Monitoring and Alert System | Arduino Uno, Ultrasonic Sensor, GSM Module

- Designed IoT-based medical monitoring system tracking saline levels with 2cm precision using ultrasonic sensors, triggering automated alerts to prevent 100% of empty-bottle incidents during 50-hour testing
- Programmed dual-alert mechanism combining buzzer notifications and GSM-based SMS to 3 medical staff contacts, reducing response time by 75% in clinical simulation environments
- Reduced patient monitoring workload by 40% for nursing staff through real-time automated surveillance

Skills

Programming Languages: Python | C | C++ | SQL | MATLAB | Arduino C
ML & Data Science: TensorFlow | Keras | Scikit-learn | NumPy | Pandas | OpenCV | YOLOv5 | CNN
Engineering Software: ETAP | ANSYS | Power BI | Arduino IDE | VS Code | Git | Jupyter Notebook
Power Systems: Substation Operations | Protection Systems | HVDC Transmission | Power Analysis
Industrial Automation: Motor Control | VFD Systems | PLC Programming | SCADA Monitoring
Core Technical: Computer Vision | IoT Development | Embedded Systems | OOP | DSA

Work experience

Tech Solutions Inc. | ML Intern *Kolkata, India | May 2025 – July 2025*

- Automated data pipelines for 5000+ records, cut preparation time by 60%
- Optimized 8 ML scripts, boosted prediction speed by 35%
- Tested models with 4-person team, hit 92% accuracy across 3 use cases

Steel Authority of India Limited (SAIL) | Internship Trainee *Bokaro, India | Nov 2024 – Dec 2024*

- Maintained 15+ motors (5-50HP), achieved 85% equipment uptime
- Documented procedures for 12 machines, adopted by 3-person team
- Supported daily safety and start-up checks on key equipment lines

Damodar Valley Corporation (DVC) | Internship Trainee *Mejia, India | June 2024 – July 2024*

- Analyzed 132kV substations serving 50MW+ power network
- Observed daily operation routines and fault-handling steps with senior engineers

Publications & Research

○ **"Characteristics Investigation of Dual Rotor Single Stator 6-Phase Single I-Shaped Magnetic Pole PMSM for Smart Electric Vehicles" |** IEEE GIEST 2024 Conference | Co-authored research analyzing electromagnetic characteristics of novel PMSM topology for EV applications

○ **"Noise Filtration and Voice Discrimination: Case Studies, Results, and Discussions" |** Journal Publication Documented methodologies for audio preprocessing and voice classification with comparative analysis of 5+ filtering techniques

Certifications & Awards

○ **Best Paper Award – IEEE GIEST 2024 Conference (Recognized among 150+ submissions for research excellence in electric vehicle motor design)**

○ **100 Days of Code: The Complete Python Pro Bootcamp – Udemy (Certificate of Completion, 2024)**